



Software Technology Department

Course Code : CCPROG1
 Course Title : Logic Formulation and Introductory Programming
 College / Department : College of Computer Studies / Software Technology
 Type of Course : Foundation course
 Prerequisites :
 Pre-requisite to : CCPROG2 (Advanced Programming)
 Term / Academic Year : Term 1, AY 2024 – 2025
 Class Schedules and Instructors :

Section	Online Day	Time	F2F Day	Time	F2F Venue	Instructor
S15B	M	0730 - 0900	H	0730 - 0900	G306B	Jackylyn Beredo Communication: via AnimoSpace inbox Consultation hours: M 0900 - 1200 by appointment
S17B	M	0730 - 0900	H	0915 - 1045	G302B	
S19A	M	1430 - 1600	H	1245 - 1415	G302A	
S28	M	1430 - 1600	H	1430 - 1600	G302A	
S21A	T	1430 - 1600	F	1245-1415	G302A	Dr. Jennifer Contreras Communication: via jennifer.contreras@dlsu.edu.ph Consultation hours: MW 1100 - 1300 by appointment
S21B	T	1430 - 1600	F	1430-1600	G302A	
S18A	M	1615 - 1745	H	1615 - 1745	G302A	Andrian Jim Cubillas Communication: via AnimoSpace Inbox Consultation hours: W 11:00 - 13:00, 18:00 - 21:00 with prior appointment
S29	M	1615 - 1745	H	1800 - 1930	G302A	
S30	T	1800 - 1930	F	1800 - 1930	G302A	
S15A	M	0730 – 0900	H	0730 – 0900	G302A	Nathalie Rose Lim-Cheng Communication: strictly via nathalie.lim@dlsu.edu.ph Consultation hours: M 0900 – 1200 and W 0800 – 1200, with prior appointment
S17A	M	0730 – 0900	H	0915 - 1045	G302A	
S22A	T	0730 – 0900	F	0730 – 0900	G302A	
S23A	T	0730 – 0900	F	0915 - 1045	G302A	
S18B	M	1615 - 1745	H	1615 - 1745	G302B	Louis Patrice Lu Communication: via louis.lu@dlsu.edu.ph Consultation hours: by appointment
S19B	M	1615 - 1745	H	1430 - 1600	G302B	
S12B	M	0915 - 1045	H	0730 - 0900	G302B	Diosdado Nicdao Communication: via diosdado.nicdao@dlsu.edu.ph Consultation hours: WF 11:00-13:00, with prior appointment
S16A	M	0915 - 1045	H	0915 - 1045	G210	
S26B	T	0915 - 1045	F	0915 - 1045	G306B	John Peter Ramos Communication: via john.peter.ramos@dlsu.edu.ph Consultation hours: By appointment
S27B	T	0915 - 1045	F	1100 - 1230	G304A	
S24B	T	0730-0900	H	0730-0900	G210	Florante R. Salvador Communication: via florante.salvador@dlsu.edu.ph Consultation hours: Online T 1045-1200 In-person F 1045-1200
S13A	M	1100 - 1230	H	1100 - 1230	G302A	John Alexander Santillana strictly via Google Chat
S14B	M	1100 - 1230	H	0915 - 1045	G306B	
S24A	T	1100 - 1230	F	1100 - 1230	G302A	

S25A	T	1100 - 1230	F	0915 - 1045	G306A	Consultation hours: MT 1400 - 1800, W 1400 - 1700 and H 1400 - 1700 by appointment
S25B	T	1100 - 1230	F	1430 - 1600	G306B	
S13B	M	0730 – 0900	H	0730 – 0900	G304A	John Byron Tuazon Communication: via john.byron.tuazon@dlsu.edu.ph Consultation hours: Monday: 09:00 to 14:40 (via john.byron.tuazon@dlsu.edu.ph) Tuesday: 09:00 - 10:30 (via john.byron.tuazon@dlsu.edu.ph) Thursday: 11:00 - 14:30 (in campus) by appointment
S20B	M	0730 – 0900	H	0915 - 1045	G304A	
S22B	T	0730 – 0900	F	0730 – 0900	G302B	
S23B	T	0730 – 0900	F	0915 - 1045	G302B	
S12A	M	0915 - 1045	H	0730 - 0900	G306A	
S14A	M	0915 - 1045	H	0915 - 1045	G306A	Gary Soriano Communication: via gary.soriano@dlsu.edu.ph Consultation hours: MT 0730 - 0900 (by appointment)
S11A	T	0915 - 1045	F	0915 - 1045	G210	
S27A	T	0915 - 1045	F	0730 - 0900	G304A	
S11B	T	0915-1045	F	0915-1045	G211	Leif Romeritch Syliongka Communication: via Canvas Inbox Consultation hours: by appointment
S16B	M	0915-1045	H	0915-1045	G211	
S20A	M	0915-1045	H	1100-1230	G304A	
S26A	T	0915-1045	F	1100-1230	G306A	

Course Site / Repository : AnimoSpace <https://animospace.dlsu.edu.ph>
 Course Group : AnimoSpace Course: **CCPROG1**
 Estimated time for study outside class : 4.5 hours per week

Course Description:

This course covers the fundamentals of logic formulation together with their implementation in the C programming language. This course serves as a foundation for future courses that the students will encounter throughout their program.

Learning Outcomes (LO):

At the end of the course, the student is expected to be able to do the following¹:

Expected Lasallian Graduate Attributes	Learning Outcomes
Critical and Creative Thinker, Reflective Lifelong Learner	LO1. Analyze problem requirements by describing input specifications, processes and target output [CS01.03]
Critical and Creative Thinker, Reflective Lifelong Learner	LO2. Design and implement algorithmic solutions from defined problems and requirements by applying knowledge of computing fundamentals using appropriate data types and constructs including expressions, conditional statements, iterative statements and structured decomposition. [CS01.03; CS03.01]
Critical and Creative Thinker, Effective Communicator	LO3. Design, execute and document various classes of test cases and their corresponding results. [CS05.01; CS07.02]
Critical and Creative Thinker	LO4. Determine and apply proper debugging techniques using programming constructs and/or computing tools. [CS03.01]
Effective Communicator	LO5. Apply simple coding techniques such as inline comments, version documentation, and following coding standards for program readability. [CS05.01]
Service-Driven Citizen	LO6. Exhibit intellectual honesty, responsibility and punctuality, conforming to Christian principles. [CS09.02]

¹*This footer should be about the PIs or the PGOs we are mapping to

Major Course Outputs (MCO):

Students are to develop, individually, a C application (referred to as a machine project [MP]) as one of the evidences in attaining the above learning outcomes. The detailed specifications and deliverables of this course project requirement will be provided in a separate document. The rubric is to be used as a general guideline for evaluating this project.

Learning Outcome	Required Output	Due Date
LO1- LO6	Machine Project (deliverables include source code, internal documentation, and test script)	November 25, 2024 (M) 0800

Rubric for Assessment:

Criteria	Exemplary	Satisfactory	Developing	Beginning
Program Correctness	41-75 The application meets all the requirements specified in the project specification. The code is syntactically and logically correct for all cases. Implementation of the program follows the indicated guidelines and does not violate indicated restrictions. The implementation also exhibits appropriate use of programming constructs.	26-40 The code works for typical input, but fails for minor special cases; the major requirements are met, though some minor ones are not. Some implementation of the program violates indicated restrictions.	11-25 The code sometimes fails for typical input. Many parts of the program implementation violate indicated restrictions and some parts of the solution are not implemented using appropriate programming constructs.	0-10 The code often fails, even for typical input. Most indicated restrictions were violated. <i>Note: Program that does not run and /or implemented incorrectly (based on specifications and restrictions) automatically gets 0 for this course output.</i>
Concept Understanding	4-5 Answers to questions are correct, reasonable, and reflective of the code. The justifications provided are sound.	3 Answers to questions are correct, but some justifications provided are weak.	2 Answers to questions are correct, but cannot justify solution (e.g., solution via trial and error, rather than proper understanding and application of concepts).	0-1 Correct understanding of the problem, but was unable to explain workings of code provided. <i>Note: Failure to explain and justify workings of the code submitted will automatically merit 0 for this course output.</i>
Readability	9-10 The program conforms to a coding standard that promotes code readability. Internal documentation is comprehensive. <i>Note: See references for some relevant</i>	6-8 Minor code formatting does not exhibit consistency in coding standard. Not all functions / program features have proper internal documentation.	1-5 Minimal internal documentation and code readability.	0 No internal documentation and code is not readable.

	<i>information on code readability.</i>			
Test Cases	9-10 All test cases are indicated correctly and completely with correctly formed descriptions.	6-8 All test cases are indicated correctly but some results are incomplete or missing, or some descriptions are general, or some cases are missing.	1-5 Test cases are indicated but most cases are missing, or results of most cases are missing, or descriptions are too general.	0 No test case document submitted.

Note: The maximum score in this rubric for the final MP deliverable is 100 points. An additional 10 points maximum may be given depending on the additional output outlined in the MP specs document. These additional outputs may require advanced reading and understanding of topics that are not covered in the course.

Other Requirements and Assessments:

The course has the following requirements on top of the major course output described above:

This course has one other summative assessment: a cumulative final exam. In addition, there are formative assessments such as exams and graded course activities. All these assessments cover the understanding of concepts, analysis of code, identifying test cases, and programming. Graded course activities include (but not limited to) recitation, hands-on exercises, seatwork, and assignments.

Grading System:

To pass this course, a student must earn **at least 60%** through the course requirements. Grade breakdown as well as GPA equivalents, are given below:

Requirement/Assessment Task	Weight
Midterm Exam	20%
1 Departmental Final Exam	20%
2 Hands-on Exam	20%
Machine Project	20%
Course Activities	20%
TOTAL:	100%

Min. Raw %	GPA
94	4.0
89	3.5
83	3.0
78	2.5

Min. Raw %	GPA
72	2.0
66	1.5
60	1.0
0	0.0

Teaching Methods / Strategies:

1. Lectures
2. Recitation and Short Quizzes
3. Reading Assignments
4. Exercises
5. Machine Project
6. Laboratory sessions
7. Hands on exam
8. Assessment

Learning Plan:

To achieve the learning outcomes of the course, the following learning activities, in line with the indicated topics, are prescribed.

Learning Outcomes	Topic	Week / Dates	Learning Activities
LO6	Course Requirements	1	• Explain course requirements and policies. • Discuss the importance of programming and the C language.

LO1 – LO2	The Process and Importance of Programming Problem Analysis, Logic Formulation		• Discuss the process of programming, about programming environments, compilation and execution. • Demonstrate a sample readable C programming code following some coding standard, and its execution. • Read about steps in program development and the history of C language. • <u>Edit, compile, link, execute and debug C programs using a command line interpreter (CLI) and an Integrated Development Environment (IDE)^{2*}.</u>
LO1 – LO6	Variables, Operators, Expressions, and the Assignment Statement	2	• Discussion includes: - Literals and variables - Declaring and initializing variables - Assignment operator and statement - Arithmetic operators - Parentheses, operator precedence, and operator associativity • Exercises include (but not limited to): - Evaluation of arithmetic expressions - Conversion of mathematical expressions into C expressions - <u>Observing coding standards and documentation practices advocated in the course[*].</u>
LO1 – LO6	Functions and Programs	3 – 5	• Discussion includes: - The output function, printf() - The input function, scanf() - The main function, main() - The compiler directive, #include • Exercises include (but not limited to): - Analysis of the problem by identifying the given input and the necessary formula / computation to produce expected output(s) - <u>Writing programs exhibiting the use of functions in the input / output library, use of variables, assigning values to variables, and incorporating expressions for evaluation.[*]</u> - <u>Reviewing code to identify vulnerabilities and suggest alternative solutions.[*]</u>
			• Discussion includes: - Purpose of using and creating user-defined functions - Function definition (includes function call and prototyping) - Passing parameters (of values and addresses) • Exercises include (but not limited to): - Analysis of the problem by identifying what parts / features can be implemented as subroutine(s), as well as identifying parameters and returned values, as appropriate. - <u>Identify test cases to evaluate and validate correctness of each user-defined function.[*]</u>

^{2*} **This is a recurring activity throughout the entire course.**

	Conditional Constructs	6 – 8	• Discussion includes: - Relational and equality operators - Logical operators - Conditional constructs (inclusive of nesting) • Exercises include (but not limited to): - Analysis of the problem by identifying what conditional construct best fits the problem. - Identifying appropriate operators in constructing conditions given requirements to solve programming problems. - Writing functions / programs exhibiting the use of conditional constructs.
Hands on Exam 1 – October 16, 2024 (W) shift 1: 0800 – 1030; shift 2: 1100 - 1330 (coverage until Functions)			
	Iterative Constructs	8 – 13	• Discussion of iterative constructs (inclusive of nesting) • Exercises include (but not limited to): - Analysis of the problem by identifying what iterative construct best fits the problem. - Identifying appropriate operators in constructing conditions given requirements to solve programming problems. - Writing functions / programs exhibiting the use of iterative constructs.
Midterm Exam – November 13, 2024 (W) shift 1: 0730 - 0930; shift 2: 1000 - 1200 (coverage until Simple Loops)			
Hands on Exam 2 – November 20, 2024 (W) shift 1: 0800 – 1030; shift 2: 1100 - 1330 (coverage until Nested Loops)			
Machine Project Due Date : November 25, 2024 (M) 0800			
Final Examination – December 2, 2024 (M) shift 1: 0730 - 1030; shift 2: 1100 - 1400 (tentative)			

Text / Materials / References:

Main References:

- Ruiz, C.R. (2018). **Logic Formulation and C Programming. A Course Material for CCPROG1** (available in AnimoSpace).
- Lim, N. R. (2011) **Fundamentals of C Programming. A course material in CMPPRO1**
- Supplement lecture materials are available in: <https://dlsu.instructure.com>

Other References:

- Chu, S. B. (2018). **Labor1 for CCPROG1** (available in AnimoSpace).
- Ruiz, C. R. (2015). **Laboratory Activity Manual for COMPRO1.**
- Hanly, J., & Koffman, E. (2017). **Problem Solving and Program Design in C**, 7th ed. Addison-Wesley.
- Kelley, Al.& Pohl, I. (2001). **C by Dissection: The Essentials of C Programming**, 4th ed. Addison-Wesley.
- Kernighan, B. & Ritchie, D. (2002). **The C Programming Language**, 2nd ed. Prentice Hall
- Forouzan , B. & Gilberg, R. (2006). **Computer Science: A Structured Programming Approach Using C**, 3rd ed. Course Technology.

On Code Readability (but not limited to the following):

- Nakashian, Ashod. (2011). Why Code Readability Matters. Available in <http://blog.ashodnakashian.com/2011/03/code-readability/>
- Allain, Alex. Programming Style: Writing for Readability. Available in: <http://www.cprogramming.com/tutorial/programming-style-readability.html>
- (Edited) Zhang, Andy et al. How to Improve the Readability of Your Software Code. Available in: <http://www.wikihow.com/Improve-the-Readability-of-Your-Software-Code>

- Guzel, Burak. (2011). Top 15+ Best Practices for Writing Super Readable Code. Available in: <http://net.tutsplus.com/tutorials/html-css-techniques/top-15-best-practices-for-writing-super-readable-code/>
- Sonmez, John. (2013). What Makes Code Readable: Not What You Think. Available in: <http://www.simpleprogrammer.com/2013/04/14/what-makes-code-readable-not-what-you-think/>

Software Tools:

- Text Editor/IDE for Windows Users
 - Notepad++ (<http://notepad-plus.sourceforge.net/uk/site.htm>)
 - DevC++ (<https://sourceforge.net/projects/orwelldvcpp/>)
- Text Editor/IDE for MacOS Users
 - XCode
 - Terminal

Web References:

- Free Software Foundation. (2008). The GNU C Reference Manual. Free Software Foundation, Inc. Available in: <http://www.gnu.org/software/gnu-c-manual/gnu-c-manual.html>

General Course Policies:

- All University policies on plagiarism, cheating, proper [online and face-to-face] decorum, among others will be enforced. Refer to the Student Handbook for the details of these policies, as well as the memo from the Provost.
- Aside from these policies, standard Course policies as enumerated below will also be enforced. Additional Class policies may be provided by the instructors handling the course.
- Students are expected to read the course material prior to the discussion of the topic.
- Students are expected to answer the chapter exercises and lab manual, even when it is not assigned as homework.
- **Students are expected to attend classes on time and to participate in class activities.**
- When using the intellectual property of others (such as the use of images, source code, tools, and other software), please be reminded to properly cite your sources.
- Students are **not allowed to use, even in part, any Generative AI tool** in any course requirement or assessment, including exercises and assignments.
- It should be noted that **online sessions will be recorded**. These may be delivered via BigBlueButton, Google Meet or Zoom. You are required to **use your official DLSU email account** to access these. For Google Meet and Zoom, meeting code/link will be provided via announcements over **AnimoSpace**.
- Materials made available to the class (lectures, exercises, slides, recordings, etc.) are only for the sole use of the students enrolled in this course this term. Unauthorized screen captures, video captures, and/or distribution/sharing of materials are **NOT allowed**.
- Be mindful on what you post in the chat messages or discussions pages to be considerate to others (i.e., show respect for other's opinions) and to only post comments/questions pertinent to the discussion.
- All assignments must be submitted **BEFORE the deadline**.
- Not following instructions in the requirements for submission may result to non-acceptance of the submission (and thus, it is considered a missed exercise).
- **Late submissions will NOT be accepted.**
- Missed activities and deliverables (e.g. cases, exercises, short quizzes, projects) **CANNOT** be made up.
- The student who was absent in previous meeting/s is expected to catch up on the missed discussion himself/herself (by asking his/her peers what he/she had missed). Recordings of the online session lectures are also made available for review.
- All exams are administered **face-to-face in campus** and are individual and **closed notes**.
- **No special projects will be given.** Final grade computation is based on the same set of activities and deliverables as expected from the entire class.
- Concerns and/or disputes on results of assessments should be raised by the student on the day of the release of results. After which, all results are considered final.
- Only valid and verifiable reasons for missed examinations will be considered for make-up.
- Comply with Section 4.15 Policy on Academic Honesty of the [DLSU Student Handbook \(Links to an external site.\)](#).
- The information and data contained in the online learning modules, such as the content, audio/visual materials or artwork are considered the intellectual property of the author and shall be treated in accordance

with the IP Policies of DLSU. They are considered confidential information and intended only for the person/s or entities to which they are addressed. They are not allowed to be disclosed, distributed, lifted, or in any way reproduced without the written consent of the author/owner of the intellectual property.

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